



---

## Lab Tasks

### □ Task 1: Array Declaration and Initialization:

1. Declare an integer array called "numbers" with a size of 5.
2. Initialize the array with values 10, 20, 30, 40, and 50.
3. Display the values of the array elements using a loop.

↩ **Input:**

```
#include <iostream> // input/output library using
namespace std; // standard namespace

int main()
{
    // declare integer array of size 5 and initialize
    it
    int numbers[5] = {10, 20, 30, 40, 50};

    // loop starts from index 0 to 4
    for(int i = 0; i < 5; i++)
    {
        // display each element with its index
        cout << "numbers[" << i << "] = " <<
numbers[i] << endl;
    }

    return 0; // program ends successfully
}
```

↩ **Output:**

10  
20  
30  
40  
50

### □ Task2: Accessing Array Elements

1. Declare a character array called "message" to store a word of your choice.
2. Use a loop to access each character of the array and display it on a separate line.

#### ↗ Input:

```
#include <iostream> // input/output library
using namespace std; // standard namespace

int main()
{
    // character array (string) initialization
    char message[] = "Hello";

    // loop runs until null character '\0'
    for(int i = 0; message[i] != '\0'; i++)
    {
        // print each character on new line
        cout << message[i] << endl;
    }

    return 0; // program ends
}
```

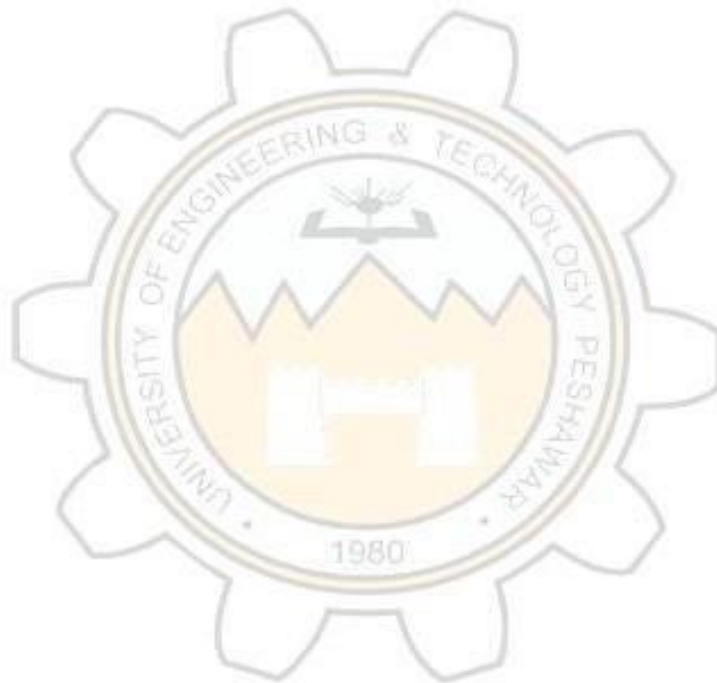
#### ↗ Output:

---

### □ Task3: CopyingArrays

1. Declare two integer arrays: "source" and "destination", each with a size of 5.
2. Initialize the "source" array with some values.
3. Use a loop to copy the elements of the "source" array into the "destination" array.
4. Display the values of both arrays to verify the copy operation.

↩ **Input:**



```

#include <iostream> // input/output library
using namespace std; // standard namespace

int main()
{
    // source array with initial values
    int source[5] = {1, 2, 3, 4, 5};

    // destination array (empty)
    int destination[5];

    // loop for copying elements
    for(int i = 0; i < 5; i++)
    {
        // copy each element from source to destination
        destination[i] = source[i];
    }

    // display source array
    cout << "Source Array: ";
    for(int i = 0; i < 5; i++)
    {
        cout << source[i] << " ";
    }

    cout << endl;

    // display destination array
    cout << "Destination Array: ";
    for(int i = 0; i < 5; i++)
    {
        cout << destination[i] << " ";
    }

    return 0; // program end
}

```

➤ **Output:**

```

Source array: 1 2 3 4 5
Destination array: 1 2 3 4 5

```

#### □ Task4:Array Sum:

1. Write a C program that prompts the user to enter 5 numbers.

2. Store the numbers in an integer array.
3. Calculate the sum of all the numbers in the array.
4. Display the calculated sum.

#### **Input:**

```
#include <iostream> // input/output library
using namespace std; // standard namespace

int main()
{
    // array to store 5 numbers
    int numbers[5];

    // variable to store sum (initially 0)
    int sum = 0;

    // loop for taking input from user
    for(int i = 0; i < 5; i++)
    {
        cout << "Enter number " << i+1 << ": ";
        cin >> numbers[i]; // store input in array
    }

    // loop to calculate sum
    for(int i = 0; i < 5; i++)
    {
        sum = sum + numbers[i]; // add each element
    }

    // display final sum
    cout << "Sum of array = " << sum << endl;

    return 0; // program end
}
```

#### **Output:**

```
Enter 5 numbers :  
5  
15  
30  
40  
10  
The sum is :100
```

---

### Conclusion:

Is lab mein hum ne C++ arrays ke basic operations seekhe:

- Array declaration aur initialization
- Character array traversal
- Array copying technique
- Loop ki madad se sum calculation

Arrays programming ka strong base hain jo data handling ko easy aur efficient banate hain.